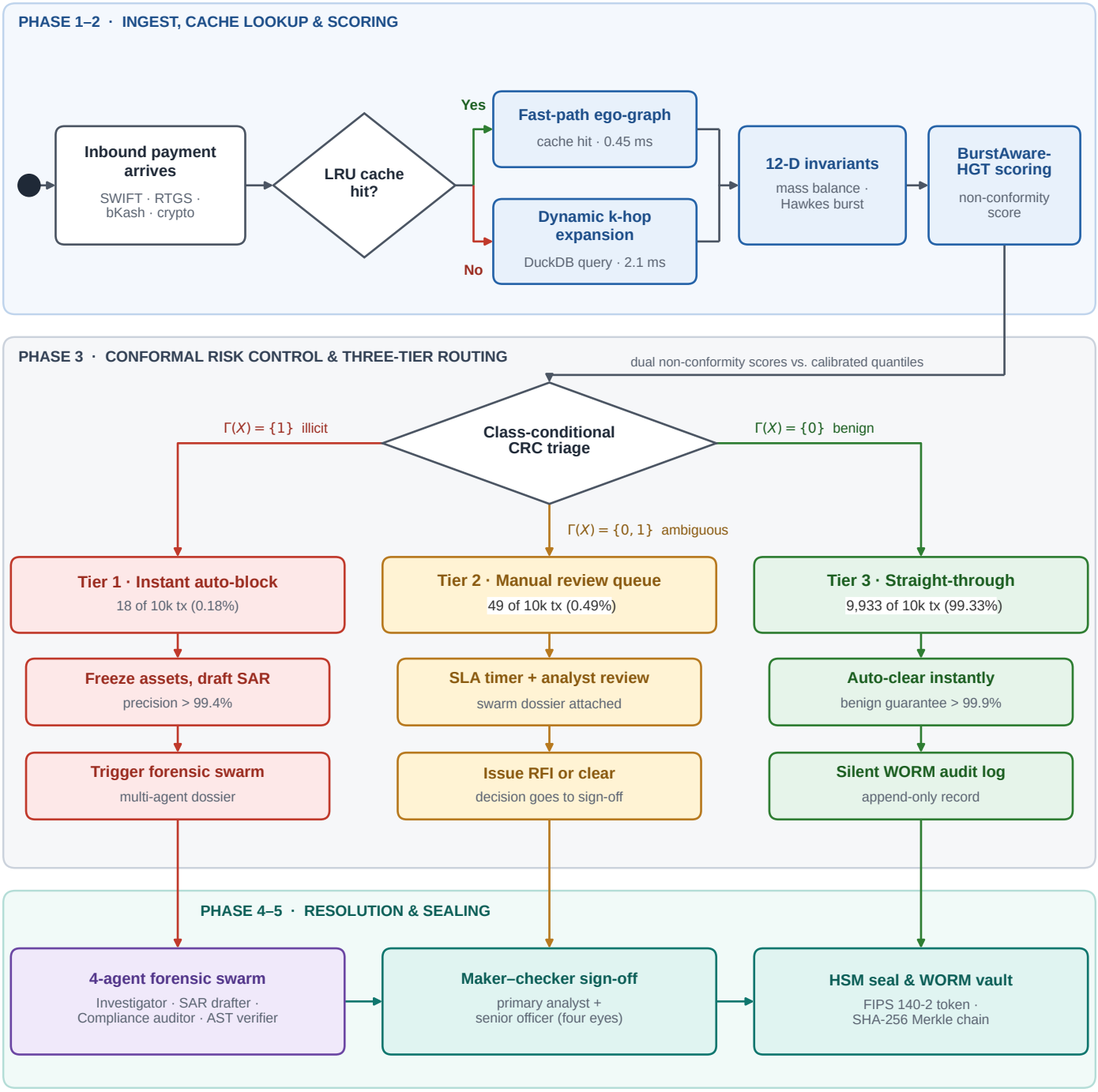
PHASE 1–2 · INGEST, CACHE LOOKUP & SCORING
Inbound payment arrives
SWIFT · RTGS · bKash · crypto
LRU cache hit?
Yes
No
Fast-path ego-graph
cache hit · 0.45 ms
Dynamic k-hop expansion
DuckDB query · 2.1 ms
12-D invariants
mass balance · Hawkes burst
BurstAware-HGT scoring
non-conformity score
PHASE 3 · CONFORMAL RISK CONTROL & THREE-TIER ROUTING
dual non-conformity scores vs. calibrated quantiles
Class-conditional CRC triage
Γ(X) = {1} illicit
Γ(X) = {0} benign
Γ(X) = {0, 1} ambiguous
Tier 1 · Instant auto-block
18 of 10k tx (0.18%)
Tier 2 · Manual review queue
49 of 10k tx (0.49%)
Tier 3 · Straight-through
9,933 of 10k tx (99.33%)
Freeze assets, draft SAR
precision > 99.4%
SLA timer + analyst review
swarm dossier attached
Auto-clear instantly
benign guarantee > 99.9%
Trigger forensic swarm
multi-agent dossier
Issue RFI or clear
decision goes to sign-off
Silent WORM audit log
append-only record
PHASE 4–5 · RESOLUTION & SEALING
4-agent forensic swarm
Investigator · SAR drafter · Compliance auditor · AST verifier
Maker–checker sign-off
primary analyst + senior officer (four eyes)
HSM seal & WORM vault
FIPS 140-2 token · SHA-256 Merkle chain